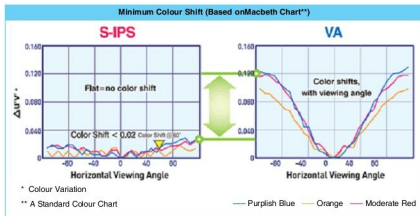
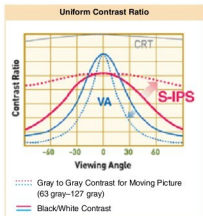


Superior Display Quality

* Little change in contrast ratio from almost any viewing angle

* Virtually no colour shift



With IPS technology, there is little change in contrast ratio when TV images are viewed from any angle

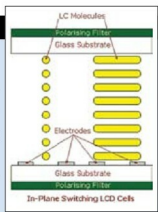
by this truly amazing technology is lesser than of the conventional vertical-alignment (VA) panel. Also, when a VA LCD TV is watched within the viewing angle, the image on the screen looks normal. But when moved away from the viewing angle, a shade of pink appears on the screen. For example, if you are viewing a rose on a VA screen at the desired viewing angle, the colour remains the same, but if you move at an angle greater than the viewing angle or move away from the viewing angle, a shade of pink is seen on the screen. However, LCD panels with IPS technology delivers consistent colours in motion pictures from all viewing angles."

Dr Sharma further states, "The horizontal alignment of liquid crystal molecules increases the stability of the LCD set, which is not the same in the conventional VA technology." In the case of VA, when the user touches the screen, the image appears blurred on the screen. This is not the case with IPS technology, explicates Dr Sharma. "Moreover, IPS has better longevity and durability than VA screens," informs Tiwari.

Power consumed by this technology is also less. "About 62 per cent less power is consumed by IPS compared to VA. But, in India, since we keep higher contrast for sharpness of the image, the power consumption is bet-

How IPS Works

IPS LCD cell has two thin glass substrates between which the liquid crystal (LC) molecules are found. LC molecules are placed parallel to each other and the glass substrate. There are electrodes in the lower glass substrate and when voltage is applied to it, a magnetic field is generated and the LC molecules arrange themselves accordingly in such a way that light can make its way and illuminate the pixels. The LC molecules can move freely at an angle of 90 degrees keeping themselves parallel to each other and the substrate. The viewing angle in LCD screens is affected by the arrangement of the LC molecules.



ter by 40 per cent," states Tiwari. After IPS technology was tested against VA, it was found that it has a maximum viewing angle of 178 degrees—much higher than VA.

Sports, games made easy

Explaining the characteristics that make IPS good for watching intensive sports and for playing PS3 games, Tiwari says, "The light rays that are emitted from the LCD screen are harmful to the eyes. Moreover, if you are playing games on a PS or watching intensive sports, the eyes get affected due to the constantly moving pictures. But in the new IPS technology, the light rays that are emitted from the screen do not affect the eyes to the extent the others do. This is because the temperature of IPS is better than

of other technologies."

IPS is robust. Although this does not give any benefit to the consumers, the manufacturer and the retailer benefit from it. "In the showrooms, where the products are showcased, the sales person may not be very educated, or may not know the proper technique of cleaning the LCD lens. In such cases, he might put scratches on the screen while cleaning. The IPS technology has advantage—it is scratchproof, whereas the conventional ones are not," explains Tiwari.

LG has already launched the first series and the second series by April end. Moreover, except for the 48cm series, all the other series will have the IPS technology. LG sold around 160,000 LCD TVs in 2008 and hopes to take the figure to 500,000 units this year. ●